Tutorial: Carrier Diffusion and Non-Equilibrium Recombination

(Direct/Radiative, Trap-Assisted SRH, and Auger)

Solid State Physics Lessons

November 5, 2025

Useful data (unless otherwise stated)

- Elementary charge $q = 1.602 \times 10^{-19}\,\mathrm{C}$, $\quad kT/q \approx 0.0259\,\mathrm{V}$ at $300\,\mathrm{K}$
- Silicon (Si, $300\,\mathrm{K}$): $n_i \approx 1.0 \times 10^{10}\,\mathrm{cm}^{-3}$, typical mobilities $\mu_n = 1350\,\mathrm{cm}^2/(\mathrm{V\,s})$, $\mu_p = 480\,\mathrm{cm}^2/(\mathrm{V\,s})$.
- Einstein relation (non-degenerate, near equilibrium): $D = \mu\, kT/q$.
- Radiative recombination (direct-gap): $R_{\mathrm{rad}} = B\,(np - n_i^2)$, B e.g. GaAs $\sim 1 \times 10^{-10}\,\mathrm{cm}^3/\mathrm{s}$; Si is much smaller ($\sim 10^{-14}$–$10^{-15}$).
- SRH recombination:

$$U_{\mathrm{SRH}} = \frac{np - n_i^2}{\tau_{p0}(n + n_1) + \tau_{n0}(p + p_1)}, \qquad n_1 = n_i e^{(E_t - E_i)/kT}, \;\; p_1 = n_i e^{-(E_t - E_i)/kT}.$$

- Auger recombination:

$$R_{\mathrm{Aug}} = C_n n(np - n_i^2) + C_p p(np - n_i^2)$$

with typical Si values $C_n \sim 2.8 \times 10^{-31}$ and $C_p \sim 1.0 \times 10^{-31}$ cm^6/s.

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1 Part A — Numerical Problems (10)

Problem 1: Mixed drift–diffusion currents (Si)

Problem Statement. Given $\mu_n = 1350\,\mathrm{cm}^2/(\mathrm{V\,s})$, $\mu_p = 480\,\mathrm{cm}^2/(\mathrm{V\,s})$, $E = 20\,\mathrm{V/cm}$, $n = 5.0 \times 10^{15}\,\mathrm{cm}^{-3}$, $p = 1.0 \times 10^{14}\,\mathrm{cm}^{-3}$, $\frac{\mathrm{d}n}{\mathrm{d}x} = -5.0 \times 10^{17}\,\mathrm{cm}^{-4}$, $\frac{\mathrm{d}p}{\mathrm{d}x} = 2.0 \times 10^{17}\,\mathrm{cm}^{-4}$. Find J_n and J_p ($\mathrm{A/cm}^2$), and identify drift/diffusion parts.

Problem 2: Built-in field in a graded n-type bar

Problem Statement. An n-type Si bar has $n(x) = N_0 e^{ax}$ with $N_0 = 1 \times 10^{16}\,\mathrm{cm}^{-3}$, $a = 300\,\mathrm{cm}^{-1}$ at $300\,\mathrm{K}$. Find the electric field $E(x)$ such that $J_n = 0$ (no-current condition).

Problem 3: Diffusion length from D and τ

Problem Statement. Minority electrons in p-type Si have $D_n = 30\,\mathrm{cm^2/s}$ and lifetime $\tau_n = 5.0\,\mu\mathrm{s}$. Find the diffusion length $L_n = \sqrt{D_n \tau_n}$.

Problem 4: Turn-off transient

Problem Statement. Uniform excess carrier density $\Delta n(0) = 1.0 \times 10^{13}\,\mathrm{cm^{-3}}$ exists in p-Si. The bulk lifetime is $\tau = 4.0\,\mu\mathrm{s}$. Find (i) $\Delta n(t)$ and (ii) the time for excess to decay to 10% of its initial value.

Problem 5: Steady-state photoconductivity

Problem Statement. Uniform generation $g = 3.0 \times 10^{19}\,\mathrm{cm^{-3}s^{-1}}$ is sustained under illumination. Low-level injection; $\tau = 1.0\,\mu\mathrm{s}$; $\mu_n = 1350\,\mathrm{cm^2/(V\,s)}$, $\mu_p = 480\,\mathrm{cm^2/(V\,s)}$. Find $\Delta n = \Delta p$ and the photoconductivity change $\Delta\sigma$.

Problem 6: SRH recombination rate in n-type Si (midgap trap)

Problem Statement. n-type Si ($n_0 = 1.0 \times 10^{16}\,\mathrm{cm^{-3}}$, $n_i = 1.0 \times 10^{10}\,\mathrm{cm^{-3}}$) with a trap at midgap ($E_t = E_i$, so $n_1 = p_1 = n_i$). SRH lifetimes: $\tau_{p0} = 10\,\mu\mathrm{s}$, $\tau_{n0} = 20\,\mu\mathrm{s}$. Uniform excess $\Delta n = \Delta p = 5.0 \times 10^{13}\,\mathrm{cm^{-3}}$ (low-level). Find U_{SRH} and effective lifetime $\tau_{\mathrm{eff}} = \Delta n / U_{\mathrm{SRH}}$.

Problem 7: Radiative recombination lifetime in GaAs (high-level injection)

Problem Statement. GaAs at 300 K with radiative coefficient $B = 1.0 \times 10^{-10}\,\mathrm{cm^3/s}$. High-level injection: $\Delta n = \Delta p = 1.0 \times 10^{16}\,\mathrm{cm^{-3}}$ in intrinsic material (assume $n_i \ll \Delta n$). Find R_{rad} and $\tau_{\mathrm{rad}} = \Delta n / R_{\mathrm{rad}}$.

Solution.

Problem 8: Auger-limited lifetime in Si (high-level injection)

Problem Statement. Si at 300 K with Auger coefficients $C_n = 2.8 \times 10^{-31}\,\mathrm{cm^6/s}$, $C_p = 1.0 \times 10^{-31}\,\mathrm{cm^6/s}$. Quasi-intrinsic (equal carrier injection): $n = p = \Delta n = 5.0 \times 10^{17}\,\mathrm{cm^{-3}}$. Find R_{Aug} and $\tau_{\mathrm{Aug}} = \Delta n / R_{\mathrm{Aug}}$.

Problem 9: Effective lifetime with surface recombination

Problem Statement. Si slab thickness $W = 200\,\mu\mathrm{m}$, ambipolar diffusion coefficient $D = 15\,\mathrm{cm^2/s}$, bulk lifetime $\tau_b = 30\,\mu\mathrm{s}$. Both surfaces act as strong recombination sinks. Using first-mode approximation, find τ_{eff}:

$$\frac{1}{\tau_{\mathrm{eff}}} = \frac{1}{\tau_b} + \frac{\pi^2 D}{W^2}.$$

Problem 10: Ambipolar diffusion coefficient and transit time

Problem Statement. n-type Si: $n_0 = 5.0 \times 10^{15}\,\mathrm{cm^{-3}}$, $n_i = 1.0 \times 10^{10}\,\mathrm{cm^{-3}}$, $\mu_n = 1200\,\mathrm{cm^2/(V\,s)}$, $\mu_p = 450\,\mathrm{cm^2/(V\,s)}$. Low-level optical excitation (minority holes dominate). Find D_n, D_p, p_0, ambipolar D_a, and diffusion transit time across $W = 100\,\mu\mathrm{m}$: $t \sim W^2/(2D_a)$.

2 Part B — Conceptual Q&A (10 questions)

Q1. When is the Einstein relation $D = \mu kT/q$ valid?

Q2. Why do electrons and holes have opposite signs in diffusion current formulas?

Q3. What is the difference between low-level and high-level injection?

Q4. What do n_1 and p_1 represent in the SRH formula?

Q5. Why are deep traps (near midgap) more effective for SRH recombination than shallow traps?

Q6. When is direct (radiative) recombination the dominant loss mechanism?

Q7. How do quasi-Fermi levels relate to the mass action product np?

Q8. What is ambipolar diffusion, and when does it apply?

Q9. Why is quasi-neutrality maintained despite generation/recombination?

Q10. How does temperature affect recombination and diffusion?